Faculty of Engineering
Department of Informatics Engineering
Software Engineering and Information Systems

الجامعة السورية الخاصة
SYRIAN PRIVATE UNIVERSITY

منصة المجتمع الاكاديمي الذكي

INTELLIGENT ACADEMIC COMMUNITY PLATFORM
NexaCampusAI

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Project Importance

- Providing a unified intelligent academic environment .
- Enhance the quality of teaching and learning.
- Promoting academic interaction and knowledge sharing by strengthening communication and collaboration .
- Advancing digital education through artificial intelligence .

Project Objectives

- To **develop** an integrated intelligent academic platform.
- To establish a comprehensive educational management system .
- To enhance academic communication and research collaboration .
- To integrate **AI-powered educational services**.

LITERATURE REVIEW

- **First Literature : Moodle [1]**
- **Second Literature : Google Classroom [2]**
- **Third Literature : MicroSoft Team [3]**

LITERATURE REVIEW

Features	Moodle	Google Classroom	MicroSoft Team	NexaCampusAI
User Authentication	✓	✓	✓	✓
Student/Professor/Admin	✓	✓	✓	✓
Admin approval for Student/Professor	✗	✗	✗	✓
Course Management	✓	✓	✗	✓
Assignment & Grades	✗	✗	✗	✓
Quizzes & Exams	✓	✓	✓	✓
Academic Communities	✓	✗	✓	✓
Academic Events	✗	✗	✓	✓
Student Project Sharing	✗	✗	✓	✓
AI Academic Assistant	✗	✗	✓	✓
AI Study Planner	✗	✗	✗	✓
Research Collaboration	✗	✗	✓	✓

GENERAL REQUIREMENT

Junior Requirments Function
User Registration
Login and Logout
Session Management
Password Recovery
Profile Creation
Profile Information Update
Profile Picture Management
Profile Customization
View Member Profiles
Text Post Publishing
Post Image Attachments
Edit Own Posts
Delete Own Posts
Social Feed
Post Details
Post Likes
Post Comments
Comment Management
Member Search
Profile Post History

Senior 1 Requirments Function
Google Single Sign-On
Outgoing Connection Requests
Incoming Connection Requests
Connection Management
Following Members
Followers and Following Lists
Social Discovery
Connection Suggestions
Filtered Member Search
Post Search
Hashtag Navigation
Story Publishing
Story Viewing and Expiration
Video Posts
Multiple Media Attachments
Saved Posts
Post Sharing and Reposting
Report Inappropriate Content
Block and Unblock Members
Social Privacy Preferences

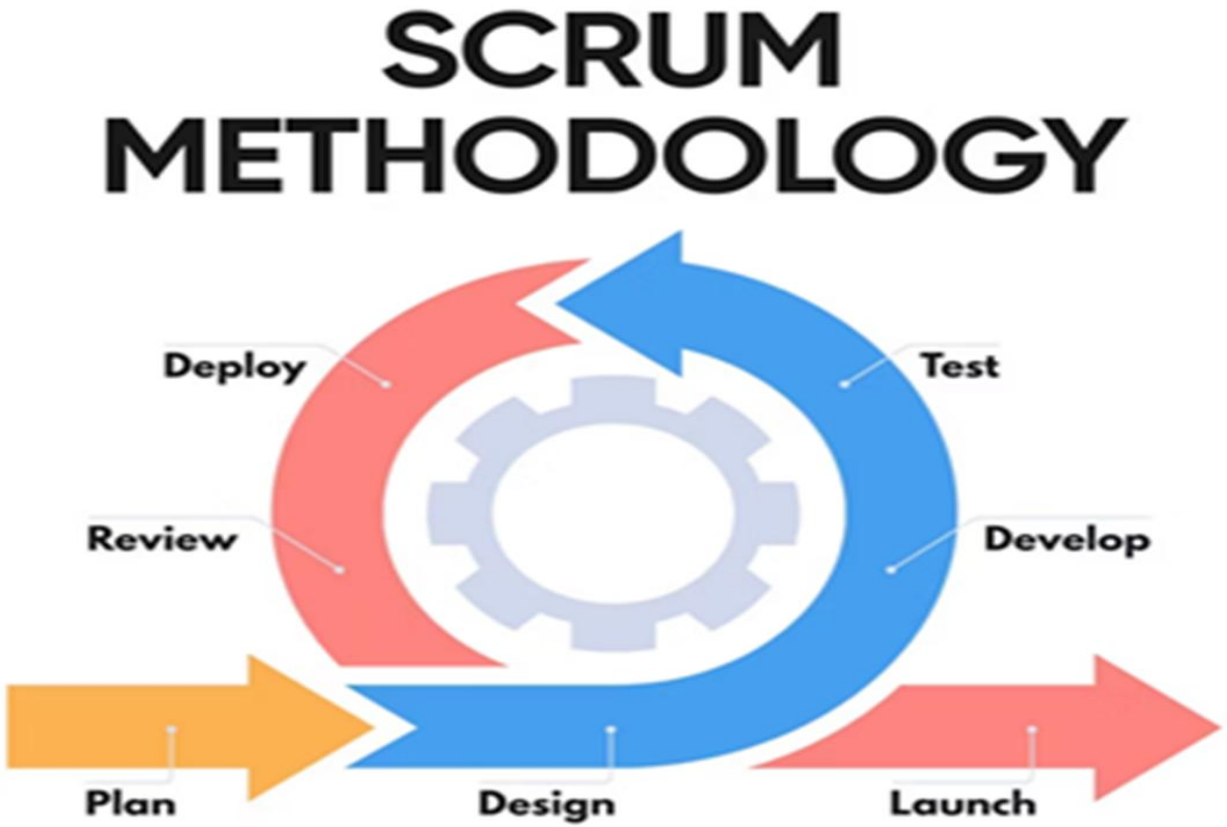
NON_FUNCTIONAL REQUIREMENT

ID	Quality attribute	Proposed requirement / acceptance direction
NFR-01	Security	The API shall validate authentication and authorization on every protected operation; cross-role and cross-faculty access attempts shall be rejected.
NFR-02	Performance	For a defined local test dataset and workload, at least 95% of ordinary non-AI interactions should complete in under 3 seconds.
NFR-03	Reliability	The system shall return actionable errors for invalid requests, database outages, and network interruptions without corrupting stored records.
NFR-04	Privacy	Personal records, submissions, protected files, and retrieved AI context shall be accessible only to authorized identities.
NFR-05	Usability	Primary student and professor tasks shall be achievable through consistent, responsive interfaces and understandable validation messages.
NFR-06	Accessibility	Critical workflows shall support keyboard navigation, labeled controls, readable contrast, and an audit against WCAG 2.2 criteria.
NFR-07	Scalability	API modules and storage shall be designed to permit future separation; multi-instance SSE requires a shared event backbone.
NFR-08	Maintainability	The project shall separate routes, controllers, models, and services and maintain documented configuration and automated test commands.
NFR-09	File Safety	Uploads shall validate type and access rights; private URLs and malware scanning require verified production configuration.
NFR-10	Backup and Recovery	Database backup and restoration shall be rehearsed in a test environment before institutional deployment.
NFR-11	Auditability	Administrative moderation, approval decisions, and critical file operations should produce traceable records with timestamps.
NFR-12	AI Answer Quality	Course-grounded responses shall be evaluated for source relevance, citation accuracy, unsupported claims, and permission leakage.

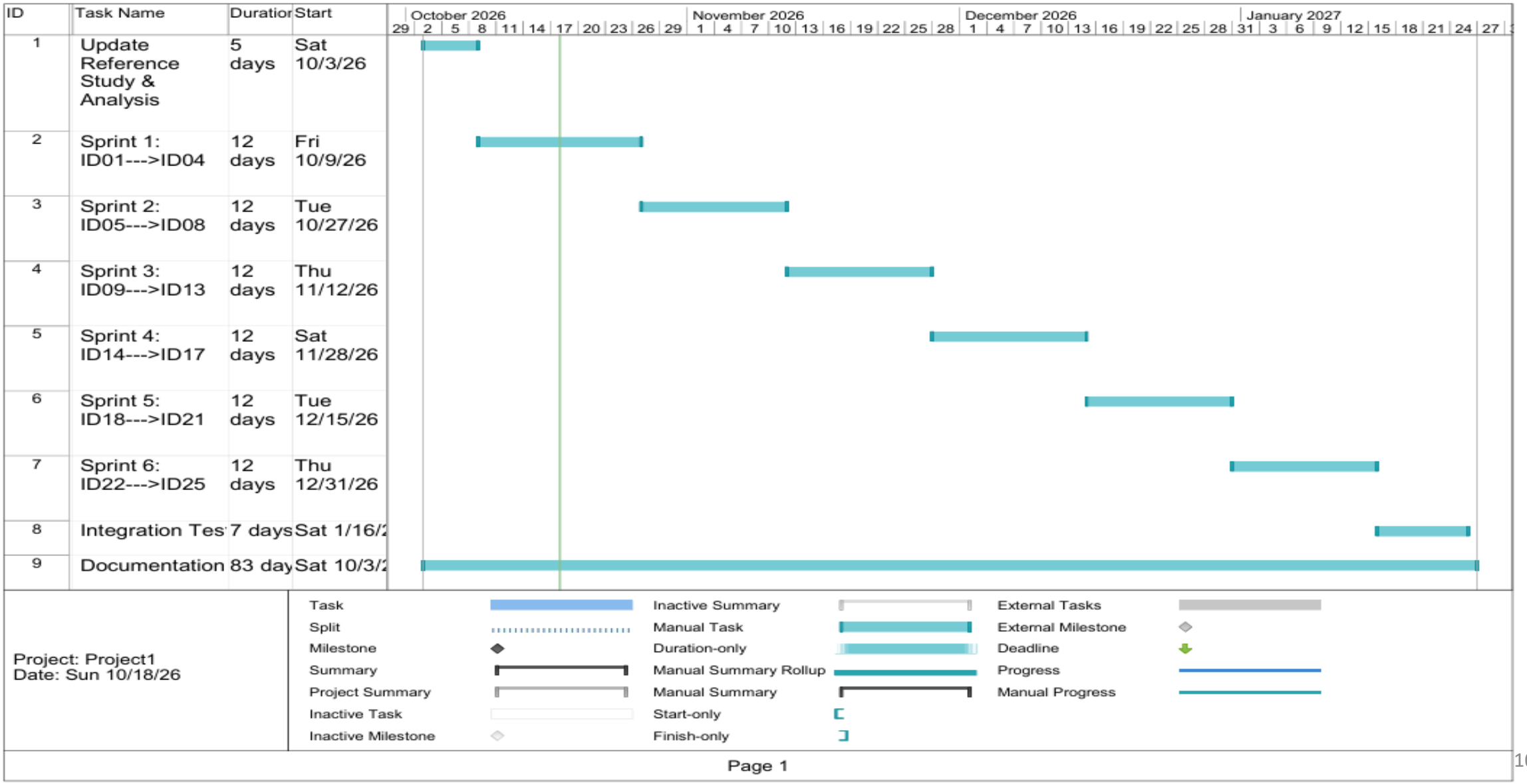
REQUIREMENTS TO BE IMPLEMENTED

ID	Functional Requirement	Priority
ID-01	Enable students and professors to submit verification requests and allow administrators to approve or reject them	1
ID-02	Control access to system features and academic information according to user roles and permissions.	1
ID-03	Automatically provide users with faculty-specific academic portals and relevant educational content.	1
ID-04	Enable professors to create and manage academic courses, including student enrollment and membership.	1
ID-05	Allow professors to organize course lectures, learning materials, and educational activities within a structured LMS workspace.	2
ID-06	Enable professors to publish assignments, receive student submissions, and provide grades and feedback.	2
ID-07	Enable professors to create, schedule, and evaluate online quizzes and examinations using reusable question banks.	2
ID-08	Allow users to upload, categorize, search, and retrieve academic resources through a centralized knowledge base.	2
ID-09	Enable authorized users to publish and manage announcements targeted to specific courses, faculties, or student groups.	3
ID-10	Allow authorized users to organize academic events and manage participant registration and event capacity.	3
ID-11	Enable users to create, join, and manage academic communities based on shared educational interests.	3
ID-12	Allow students and professors to post academic questions, provide answers, and participate in organized discussions.	3
ID-13	Enable users to exchange private messages and share attachments through real-time conversations.	3
ID-14	Allow students to publish academic projects, showcase achievements, and receive feedback from other users.	4
ID-15	Enable users to collect, organize, and format academic research references for citations and bibliographies.	4
ID-16	Allow authorized users to collaboratively create and edit academic documents while tracking document revisions.	4
ID-17	Enable professors to publish consultation availability and allow students to book available office-hour appointments.	4
ID-18	Enable Nexa AI to answer course-related questions using retrieved academic materials and provide source references.	5
ID-19	Allow students to generate AI-powered practice quizzes from selected academic topics and course materials.	5
ID-20	Generate personalized study plans and recommend relevant learning resources based on student goals and academic progress.	5
ID-21	Enable professors to analyze student performance, assessment results, and learning progress through academic dashboards.	5
ID-22	Allow professors to schedule academic activities, manage class timetables, and record student attendance.	6
ID-23	Automatically notify users about academic announcements, assignments, grades, events, and important system activities.	6
ID-24	Enable administrators to manage user accounts, review reported content, and perform academic content moderation.	6
ID-25	Allow authorized users to securely upload, store, organize, and access academic files according to access permissions.	6

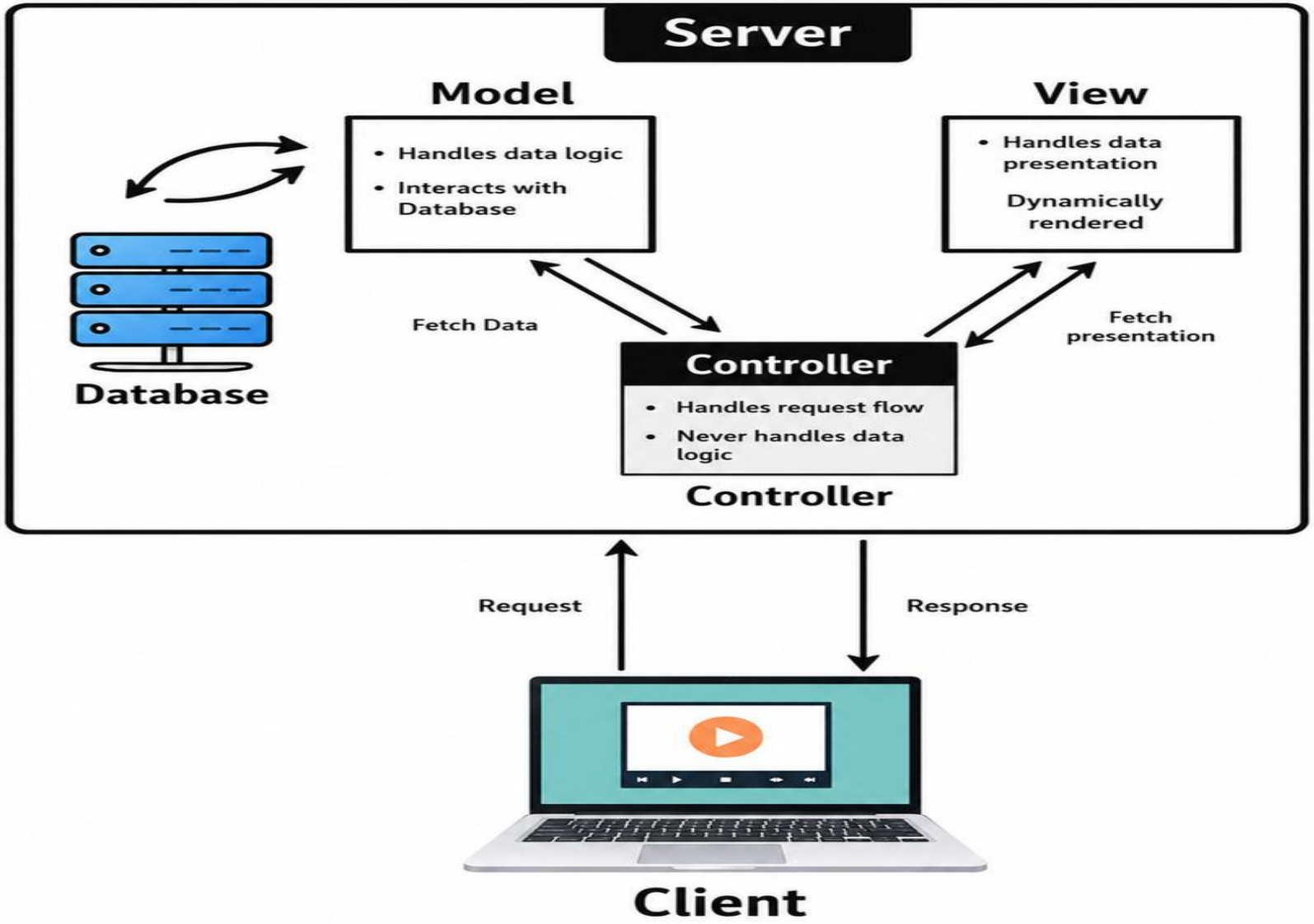
THE ADOPTED METHODOLOGY : SCRUM



Gant char (Project Timeline)



CLIENT–SERVER ARCHITECTURE AND MVC ARCHITECTURE (MODEL–VIEW–CONTROLLER)



TOOLS USED

Frontend



React.js

Backend

Node.js

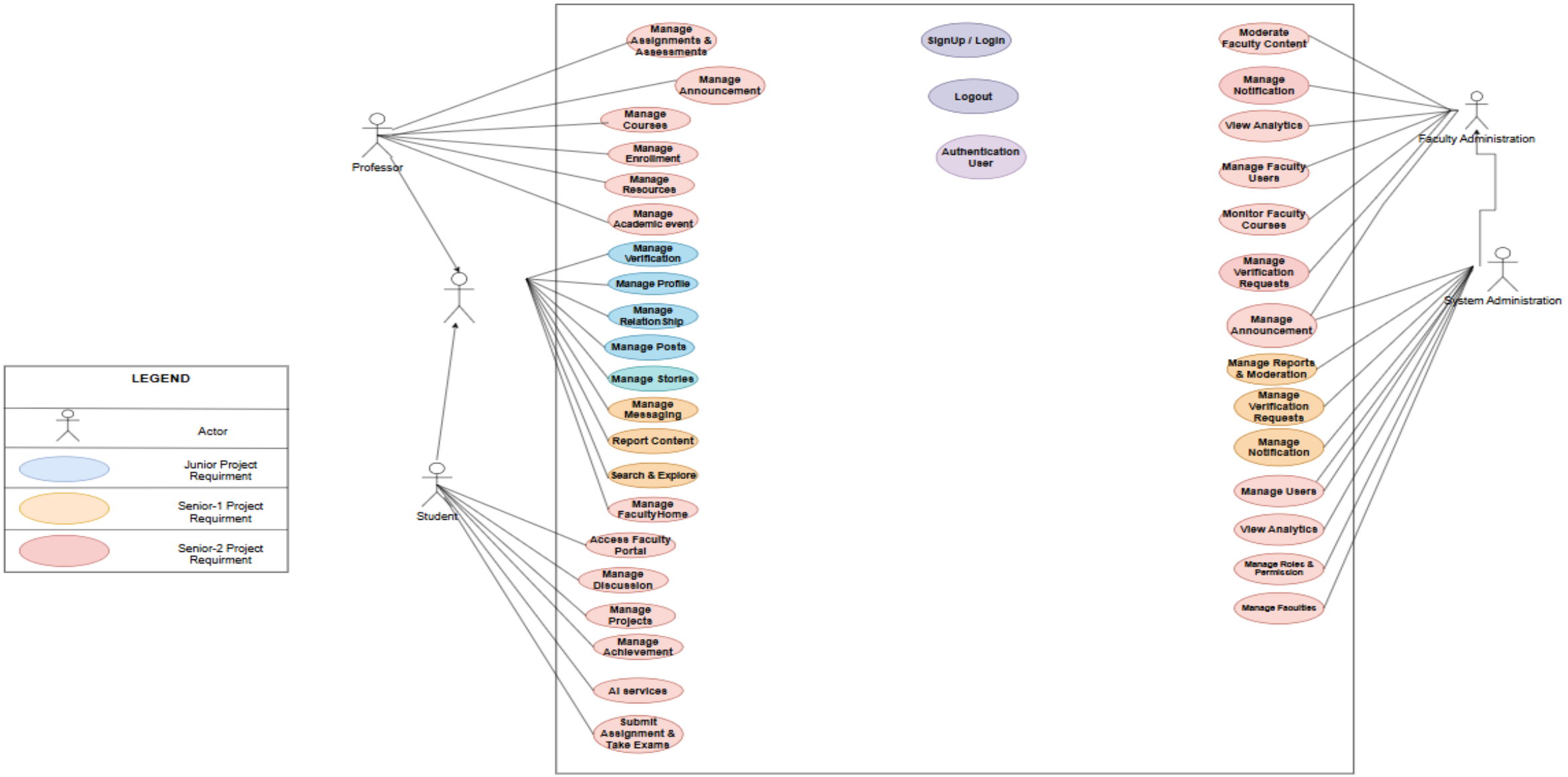
Express.js

Database



MongoDB

USE CASE DIAGRAM (HIGH_LEVEL)



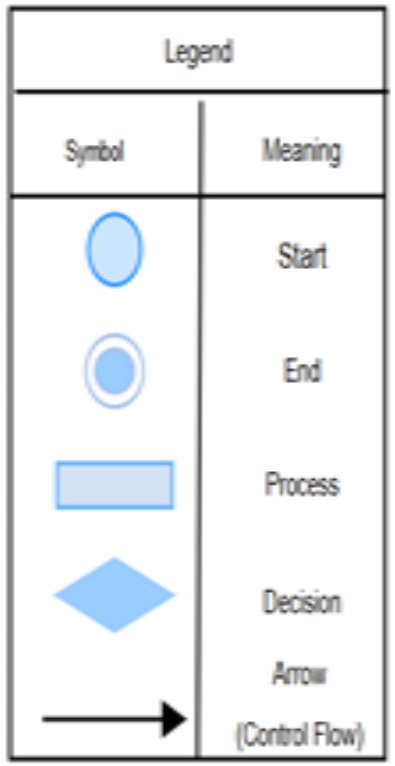
SYSTEM DIAGRAMS

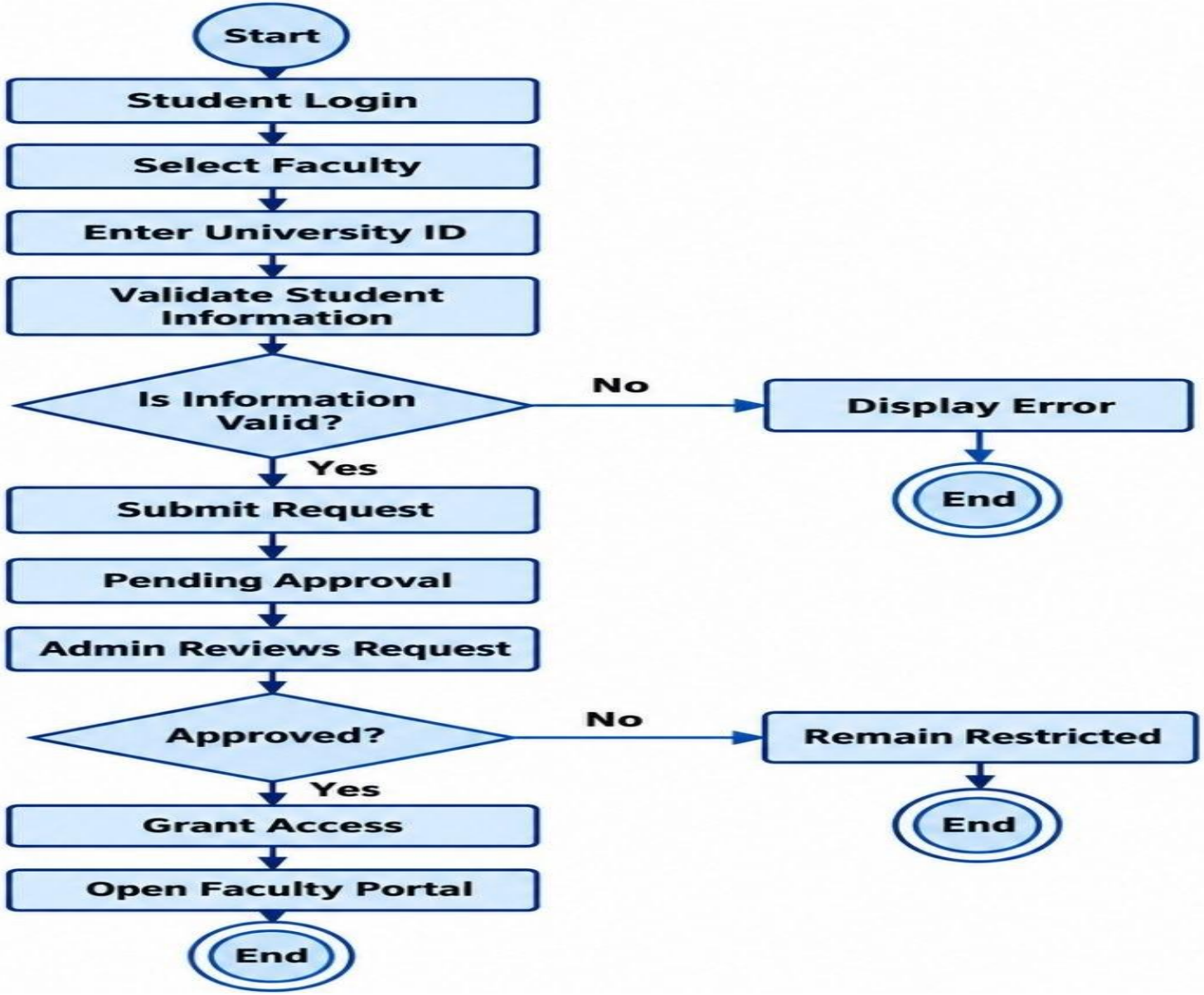
USE CASE SPECIFICATION, VERIFY STUDENT ACCOUNT (ID_01)

Field	Description
Use Case Name / ID	**Verify Student Account (ID_01)**
Actors	Student, Administrator
Description	The system verifies student identity and university information before granting access to academic services.
Preconditions	The user has authenticated and submitted valid student information.
Main Flow	1. Student signs in ,selects faculty and enters university ID. 2. System validates information. 3. Verification request is submitted. 4. . Administrator reviews the request. 5. Approved student gains access.
Alternative Flow	Invalid university ID, faculty mismatch, rejected verification, or pending approval.
Postconditions	The student account is successfully verified, the user session is created, and the student is granted access to the NexaCampusAI platform according to the assigned permissions.

SYSTEM DIAGRAMS

Activity Diagram (Verify Student Account)





SYSTEM DIAGRAMS
Activity Diagram (Verify Student Account)

